Name _____ Date _____ Period _____

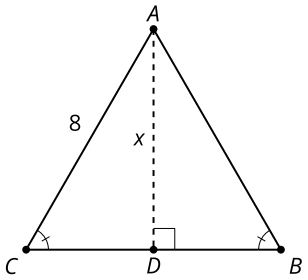
Right Triangles and Trigonometry

Geometry · Right Triangle Trigonometry

1. Triangle ABC is equilateral.

a. What is the value of x?

b. What is the measure of angle B?



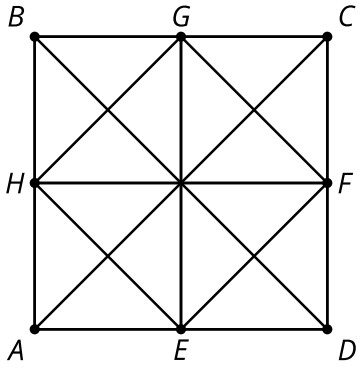
2. An equilateral triangle has side length 8 units. What is the area?

3. The quilt is made of squares with diagonals.

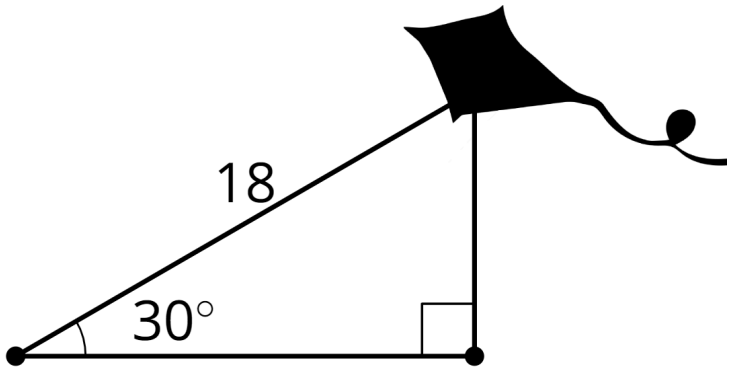
The length of BD is 4.

a. Find the length of AE.

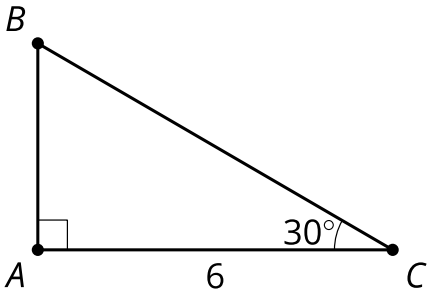
b. Find the area of square $ABCD$.



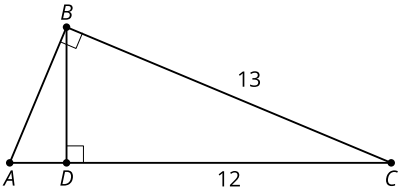
4. Kiran is flying a kite. He gets tired, so he stakes the kite into the ground. The kite is on a string that is 18 feet long and makes a 30 degree angle with the ground. How high is the kite?



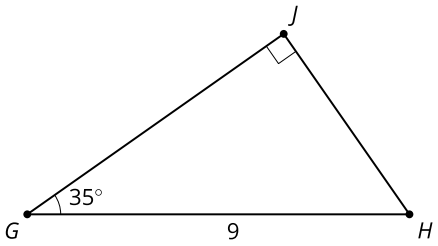
5. What is the length of side AB?



6. Find the lengths of segments AD and BD. Then check your answers using a different method.



7. Write an expression that can be used to find the length of JH and an expression that can be used to find the length of GJ.



8. A triangle has sides with lengths 5, 12, and 13.

a. Verify this is a Pythagorean triple.

b. Approximate the acute angles in this triangle.

9. Find the length of each leg.

